

4.1 Maximum and Minimum values

To start off you must read and study the definitions of a local maximum value of a function, a local minimum value of a function, an absolute maximum value of a function and an absolute minimum value of a function. These definitions can be found on page 276 of the text book.

We can interpret these definitions as follows:

$f(c)$ is a local maximum value of f if there is some open interval O containing c , with O contained in the domain of f , such that the number $f(c)$ is greater than or equal to all the function values $f(x)$ for all $x \in O$.

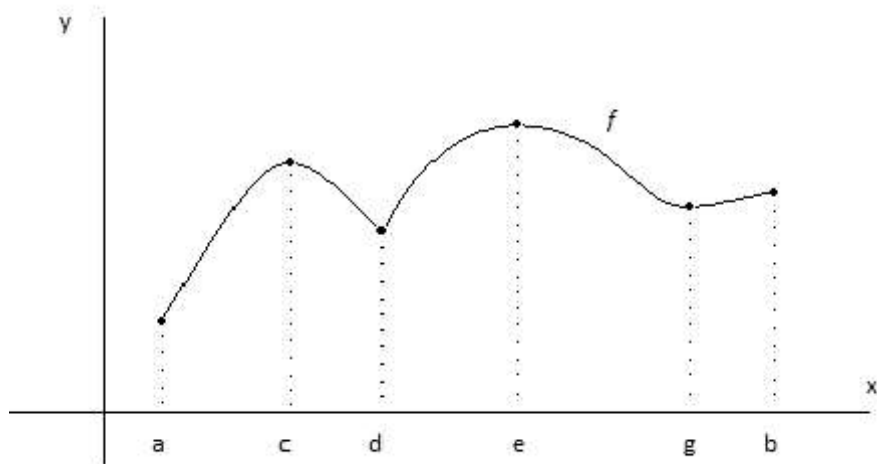
$f(c)$ is an absolute maximum value of f if the number $f(c)$ is greater than or equal to all the function values $f(x)$ for all x in the domain of f .

NOTE: If f is a function with domain a closed interval $[a, b]$, then by definition, $f(a)$ and $f(b)$ cannot be local maxima or local minima of f , since we cannot find an open interval containing a or b in $[a, b]$.

The interpretations for a local and absolute minimum value are similar.

The local and absolute maximum and minimum values of a function are called extreme values.

The following graph illustrates the difference between local and absolute extreme values:



The local minimum values are: $f(d)$ and $f(g)$. NOTE: $f(a)$ is not a local minimum since it is at an endpoint

The local maximum values are: $f(c)$ and $f(e)$. NOTE: $f(b)$ is not a local maximum since it is at an endpoint

To find the absolute minimum value of f we need to find the smallest possible value of the function on the interval $[a, b]$.

It is clear that the absolute minimum value of f is $f(a)$.

The absolute maximum value of f is $f(e)$: This is the greatest value of the function on the interval $[a, b]$.

There are two important theorems that we will use to find the extreme values of a function:

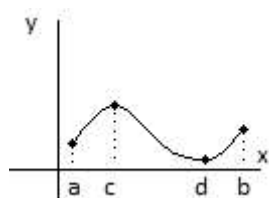
The Extreme Value Theorem (page 278) and

Fermat's Theorem (page 279)

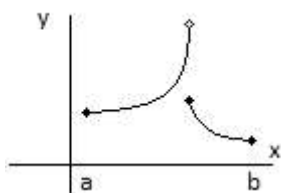
The Extreme Value Theorem tells us which functions will have extreme values:

f continuous on $[a, b] \Rightarrow$ there exists numbers $c, d \in [a, b]$ such that $f(c)$ is a absolute maximum of f on $[a, b]$ and $f(d)$ is a absolute minimum of f on $[a, b]$.

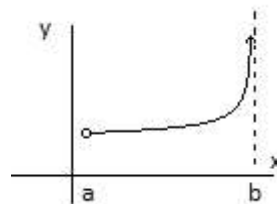
Note that you can only use this theorem if you have a continuous function on a closed interval. The following examples show you why:



1



2



3

The graph in sketch 1 shows a continuous function on the closed interval $[a, b]$ and you can see how the points c and d can be found in $[a, b]$ such that $f(c)$ is a absolute maximum of f and $f(d)$ is a absolute minimum of f .

The graph in sketch 2 shows a function that is not continuous on $[a, b]$. The function does have a absolute minimum value at b , but no absolute maximum value: The “highest” point on the graph is excluded!

The graph in sketch 3 also shows a function that is not continuous on $[a, b]$: The function is not defined at $x = a$ and also not defined at $x = b$. The function has an asymptote at b and hence there is no greatest value. Since the function is not defined at $x = a$, there is no smallest value since the “smallest” value (at a) is excluded!

Fermat’s theorem gives us a method to find the extreme values of a function. It states the following:

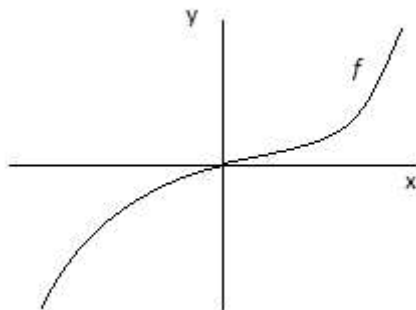
If $f(c)$ is an extreme value of f AND $f'(c)$ exists, then $f'(c) = 0$.

Be careful! The opposite is not true: If $f'(c)$ exists and $f'(c) = 0$, $f(c)$ is not necessarily an extreme value of f !

Here is an example to show this:

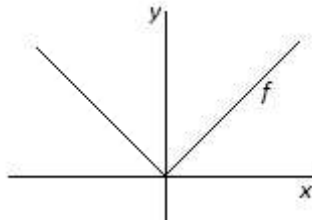
Consider the function f with $f(x) = x^3, x \in [-1, 1]$

Note that f is continuous on $[-1, 1]$ and $f'(x) = 3x^2 \Rightarrow f'(0) = 0$



However, it is clear from the graph that $f(0)$ is not an extreme value of f .

Note that a function f can have a local extreme value $f(c)$ even if $f'(c)$ does not exist, as in the following example:



Consider the function f with $f(x) = |x|$, $x \in [-1, 1]$.

Note that f is continuous on $[-1, 1]$ and it is clear that $f(0)$ is a local minimum value of f .

However, we know from previous work that $f'(0)$ does not exist.

This leads us to another concept that you must learn: A number c in the domain of a function f is called a critical number of f if either $f'(c) = 0$ or $f'(c)$ does not exist. (See definition on page 280)

Here are two examples that we have already mentioned:

1. If $f(x) = x^3$ we have seen that $f'(0) = 0$. Thus, $x = 0$ is a critical number of f .
2. If $g(x) = |x|$ we know that $g'(0)$ does not exist. Thus, $x = 0$ is a critical number of g .

Here is another example:

3. Let $h(x) = \sqrt{x}(1 - x) = x^{1/2} - x^{3/2}$

$$\text{Then } h'(x) = \left(\frac{1}{2}\right)x^{-\frac{1}{2}} - \left(\frac{3}{2}\right)x^{\frac{1}{2}} = \frac{1}{2\sqrt{x}} - \frac{3\sqrt{x}}{2}$$

It is clear that $h'(x)$ does not exist when $x = 0$.

$$\text{When } x \neq 0, h'(x) = 0 \Rightarrow \frac{1}{2\sqrt{x}} - \frac{3\sqrt{x}}{2} = 0 \Rightarrow 1 - 3x = 0 \text{ (Multiply by } 2\sqrt{x}\text{)}$$

$$\Rightarrow x = 1/3$$

Since $h'(0)$ does not exist and since $h'(1/3) = 0$, the critical numbers of h are $x = 0$ and $x = 1/3$.

SO, how do we find the extreme values of a function? The method we will use is the Closed Interval Method – you can find the details of this method on page 281.

We will use this method for the following example, which you can find on page 281:

Example 8: $f(x) = x^3 - 3x^2 + 1$, $x \in [-\frac{1}{2}, 4]$

$$\text{Note that } f'(x) = 3x^2 - 6x = 0 \Rightarrow x = 0 \text{ or } x = 2$$

Thus the critical numbers of f are $x = 0$ and $x = 2$

$$\text{Calculate the values of } f \text{ at the critical numbers: } f(0) = 1 \text{ and } f(2) = 2^3 - 3 \cdot 2^2 + 1 = -3$$

$$\text{Calculate the values of } f \text{ at the endpoints of the interval: } f\left(-\frac{1}{2}\right) = \frac{1}{8} \text{ and } f(4) = 17$$

The absolute maximum value of f is the largest of the function values at the endpoints and the function values at the critical numbers. Thus, the absolute maximum value of f is $f(4) = 17$.

The absolute minimum value of f is the smallest of the function values at the endpoints and the function values at the critical numbers. Thus, the absolute minimum value of f is $f(2) = -3$.

Here is a last example: Let $f(x) = e^x, x \in [0, 2]$

Since $f'(x) = e^x \neq 0$ for all $x \in \mathbb{R}$, it follows that f has no critical numbers.

Calculate the values at the endpoints: $f(0) = e^0 = 1$ and $f(2) = e^2$

Furthermore, since $f'(x) = e^x > 0$ for all $x \in \mathbb{R}$, it follows that f is strictly increasing.

Thus, $1 = e^0 \leq f(x) \leq f(2) = e^2$ for all $x \in [0, 2]$, and so it follows that $f(0) = 1$ is the absolute minimum value of f and $f(2) = e^2$ is the absolute maximum value of f .